

Trajectory Optimization and Pick-Up and Delivery Sequence Design for Cellular-Connected Cargo AAVs

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Abstract—In this paper, we consider a cargo autonomous aerial vehicle (AAV)-aided multi-parcel pick-up and delivery network, where the communication ability of the AAV is provided by the ground base stations (GBSs). For such a system setup, our goal is to optimize the trajectory of the cargo AAV while minimizing the combined impact of total energy consumption and total outage time. Simultaneously, we aim to maximize overall user satisfaction throughout the entire flight duration. More specifically, we propose a pick-up and delivery of AAV (PDU) framework to address this problem and this framework consists of two parts. First, a simulated annealing (SA) algorithm is used to obtain the pick-up and delivery (P&D) order of parcels. On the basis of obtaining the P&D order through SA, we further use deep reinforcement learning (DRL) to optimize the flight trajectory of the AAV to ensure the expected communication quality between the AAV and GBSs. To verify the effectiveness of our proposed algorithms, we design three baseline strategies for comparison, and also investigate the effect of using the PDU framework with different weights. Finally, numerical results show that the performance of PDU strategy is improved by about 5%-30% compared with other strategies in solving the performance tradeoff of AAV energy consumption, communication quality, and user satisfaction.

Index Terms—Cargo AAV, deep reinforcement learning, multi-parcel pick-up and delivery, simulated annealing.

I. INTRODUCTION

IN RECENT years, cellular-connected autonomous aerial vehicle (AAV) has attracted wide attention. Cellular-Connected

AAV is a new technology that has emerged in recent years, which is connected to the Internet through the cellular network to achieve remote control and real-time data exchange, which enables the ground control system to remotely monitor the status of the AAV, plan the flight path and schedule tasks, and adjust the flight strategy in real time according to environmental changes, thus improving flight safety and task execution efficiency [1]. Traditional non-networked AAVs usually rely on point-to-point communication systems, which have a short communication distance and require operators to operate within the line-of-sight (LoS) range, and cannot achieve remote control and intelligent scheduling. Therefore, the emergence of cellular-connected AAVs brings the application of AAVs into data acquisition [2], [3], [4], mobile edge computing [5], [6], [7], and security inspection [8]. Among these applications, using AAVs for delivery has been conducted in some well-known companies such as Amazon [9], JD.com [10], and SF Express [11], and they also plan to increase investment in scientific research in this field in the future to improve their service quality.

A large amount of research has proven that the application of AAVs in the field of cargo transportation has significant advantages. First, the use of AAVs has improved delivery efficiency. By utilizing cellular-connected AAVs can navigate through cities and remote areas [12]. Expanding the coverage and scope of delivery is another key benefit. AAVs are not limited by the ground infrastructure, allowing them to enter geographically challenging or isolated areas. This ability is particularly valuable for reaching areas with underdeveloped transportation networks, thereby expanding transportation range and improving accessibility [13]. Reducing costs is a significant advantage of using AAV technology in cargo delivery. Automated AAV systems can reduce operating costs and reduce reliance on manpower [14]. In addition, AAVs play a crucial role in emergencies or humanitarian assistance missions. They can quickly transport medical supplies, food, and emergency aid to difficult-to-reach areas, potentially saving lives and providing critical support during crises [15].

A. Related Works

The existing literature on the application of AAVs in the field of logistics mainly focuses on the following two aspects: 1) optimizing certain performance parameters of a single AAV for

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transporting parcels, such as total delivery time, distance, or cost, and 2) optimizing the task allocation of AAV swarms for transport and drone-truck collaborative transport. In particular, the authors in [16] considered a disconnectivity-aware AAV for transporting cargo and minimized the AAV's energy consumption and handover rate by optimizing the AAV's trajectory. Similarly, to achieve the most efficient delivery service at the lowest cost, assuming that the AAV can cover all express points, the constraints including distance, load, and time of the AAV have been taken into consideration in [17]. In [18], the authors considered a multi-AAV cooperative cargo transportation scenario, and proposed an optimal AAVs routing and loading strategy to minimize the total cost of AAVs damage, total cost of cargo damage, and expected cost of unvisited targets. In [19], the authors aimed to minimize the time required to complete a batch of orders while completing the task. In [20], taking into account the payload weight and horizontal fitness of each AAV flight in the AAV swarm, the authors applied genetic algorithms to achieve the minimum delivery time. In addition, to improve the energy efficiency and service quality, [21], [22], [23] used a drone-truck cooperative transportation system to deliver packages.

Despite the many advantages and research achievements of the above-mentioned AAVs in the cargo delivery, there are still many technical challenges. First, to enable AAVs to autonomously navigate and complete tasks, and avoid collisions or other accidents caused by communication outages, AAVs need to maintain good communication with ground base stations (GBSs) to receive GBSs flight instructions with low latency. Fortunately, some relevant works on these challenges have been studied in the literature. For example, the authors [16] stipulated that AAVs must maintain communication connection with GBSs, and further optimized the energy efficiency of AAVs. In [24], the authors minimized the task completion time of the AAV while ensuring a good LoS channel with at least one GBS at all times. In [25], the communication and task completion time of the AAV were considered. When the weight factors in the weighted sum of the communication outage time and total flight time are different, the flight strategy of AAVs also varies. In our previous work [26], we proposed a hybrid deep reinforcement learning (hybrid DRL) method to address the tradeoff between communication quality and energy consumption of the cargo AAV. Second, due to the limited battery energy storage of AAVs, they are unable to perform sustained and long-term operations. To address this challenge, the current mainstream approach is to adjust the flight strategy of AAVs reasonably to improve energy efficiency. [27] proposed an innovative algorithm that simultaneously optimized the route and perception task selection of drones to minimize energy consumption, maximize perceived returns, and ensure timely delivery of goods. On the basis of considering battery and payload weight, the authors derived and verified the energy consumption model of multi-rotor AAVs, and proposed a simulated annealing (SA) algorithm to reduce the energy consumption of AAVs [28]. In [29], the authors proposed a new collaborative pareto ant colony optimization algorithm to minimize travel costs and maximize customer service levels in terms of timely delivery. Similarly, the authors in [30] studied

a collaborative truck drone routing problem and developed an improved variable neighborhood descent (IVND) to optimize the energy efficiency of the entire system. Finally, since the delivery target is the user, user satisfaction with the service, i.e., the time required for the cargo AAV to deliver packages to the user, is an important performance indicator. The authors in [22] studied a truck drone collaborative transportation system and proposed a hybrid truck drone delivery (HTDD) algorithm to minimize the total delivery time of all packages. In [31], the authors used the hybrid DRL method to minimize the total time required for AAVs to deliver COVID-19 test kits to all users. To make readers better understand the current research status of the cargo AAV, Table I compares the current works related to the cargo AAV and the focus of these works.

B. Motivation and Contributions

From Table I, there is not any endeavor in studying a scenario in which cellular-connected AAVs are used to pick up and deliver parcels, while jointly optimizing the energy consumption, communication quality, and overall user satisfaction. In addition, there are also many constraints in the optimization problem, such as the need for AAVs to extract packages before delivering them to the corresponding users, AAVs can only communicate with one GBS during flight, and so on. Obviously, this is a highly non-convex problem, and traditional convex optimization or continuous convex approximation methods are difficult to effectively solve. At the same time, due to the existence of multiple pick-up points and delivery points, the direct use of deep reinforcement learning (DRL) algorithms lead to difficult convergence of neural network.

Motivated by this, we propose a PDU framework and divide the optimization problem into two sub-problems, i.e., the pick-up and delivery (P&D) order optimization problem of the AAV, and the path optimization problem after the P&D order is determined. To the best of our knowledge, this is the first work that simultaneously studies the tradeoff between the energy consumption of the AAV, the communication quality of the AAV, and the overall user satisfaction in the AAV logistics system. The contributions of this paper are summarized as follows:

- We propose a P&D model of the cargo AAV, in which each cargo needs to be picked up from the pick-up point and flown to the corresponding user location (delivery point) for delivery. Specifically, three factors including energy consumption, communication quality, and overall user satisfaction are jointly considered, and the objective function is expressed as the weighted sum of these three factors. Our goal is to find an optimal flight trajectory for our proposed algorithm and minimize the objective function regardless how the weights change.
- We propose a PDU framework to solve the highly-complex non-convex optimization problem and divide the problem into two sub-problems, i.e., the optimization of P&D order and the optimization of the AAV flight trajectory between any two points.
- To optimize the P&D order of the AAV, we propose a SA algorithm to obtain the optimal P&D order and the

TABLE I
SUMMARY OF THE RELATED WORKS ON CARGO AAV

Reference paper	Objective function construction			Logistics scenario		Algorithm
	Energy consumption	Communication quality	User satisfaction	Only delivery	Pick-up and delivery	
[16]	✓	✓	×	✓	×	Dynamic programming
[19]	×	×	✓	×	✓	Heuristics algorithm
[22]	✓	×	✓	×	✓	Hybrid truck-drone delivery algorithm
[27]	✓	×	✓	✓	×	Bellman-Ford
[28]	✓	×	✓	✓	×	Simulated annealing
[29]	✓	×	✓	✓	×	Collaborative Pareto Ant Colony Optimization
[30]	✓	×	✓	✓	×	Improved variable neighborhood descent
[31]	×	×	✓	✓	×	Hybrid deep reinforcement learning

speed of the cargo AAV. On the premise that the P&D strategy has been obtained and the channel environment is unknown, we use DRL to optimize the trajectory of AAV and learn a radio map based on the sampled data. To demonstrate the effectiveness of our proposed algorithm, we use three baseline strategies as benchmarks, which are User satisfaction-based strategy, Energy-based strategy, and Energy-based strategy with constant speed.

- We show the performance parameters obtained based on different strategies. The numerical results prove the superiority of PDU strategy in solving the tradeoff problems of energy consumption, communication, and overall user satisfaction in AAV delivery system, and the performance is improved by about 5%-30% compared with the comparison strategies.

C. Organization and Notations

The rest of this paper is organized as follows. Section II introduces the system model of the cargo AAV in the pick-up and delivery scenario. Section III introduces the proposed optimization problem. In Section IV, we introduce our proposed PDU framework in detail to address the non-convex optimization problem. In Section V, simulation results are presented. Finally, we summarize the conclusion in Section VI.

Notations: In this paper, scalars are represented by talic letters, while vectors are denoted by bold-face lowercase letters, and matrices are denoted by bold-face uppercase letters. Moreover, $\mathbb{R}^{N \times 1}$ denotes the space of N -dimensional real-valued vector. $\|\mathbf{a}\|$ represents the euclidean norm of vector $\mathbf{a}$. For a time-dependent function $y(t)$, $\dot{y}(t)$ denotes the first-order derivative with respect to time t .

II. SYSTEM MODEL

As shown in Fig. 1, we consider a AAV pick-up and delivery system. The AAV is flying at a speed $V_{\text{uav}}(t)$ at an altitude $H_f(t)$. The maximum flight speed of the AAV is denoted by V_m , and the fuselage weight of the AAV is represented by W_{uav} . We use a three-dimensional coordinate $\mathbf{q}_u(t) = (x_u(t), y_u(t), H_f(t)) \in \mathbb{R}^{3 \times 1}$ to represent the position of the AAV. To reduce the

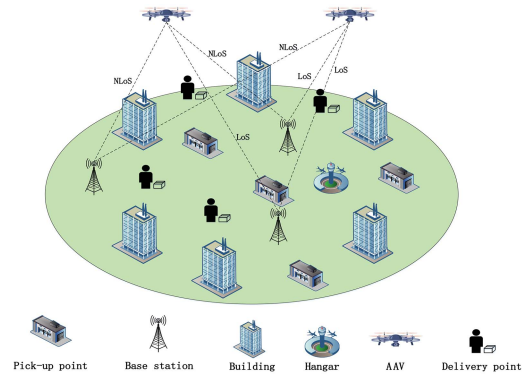


Fig. 1. System model of the cargo AAV used in a pick-up and delivery scenario

complexity of the system, this paper assumes that the AAV can ascend or descend only when it reaches the pick-up point or delivery point directly above, and flies at a constant altitude H_c at other times [32], [33].

In addition, a total of K pick-up points and K delivery points is set in the whole system, and the pick-up and delivery points are one-to-one correspondence, i.e., the parcel picked up by the AAV from the k -th pick-up point must be sent to the k -th delivery point, and the parcels that do not belong to the delivery point cannot be sent to the delivery point. We define the pick-up points with indexes from 1 to K , and the delivery points with indexes from $1'$ to K' . The set of locations of pick-up points and delivery points are denoted by $L = \{l_1, l_2, \dots, l_K\}$ and $L' = \{l_{1'}, l_{2'}, \dots, l_{K'}\}$, respectively. For any $l_k \in L$, it represents the two-dimensional coordinates (x_k, y_k) of the k -th pick-up point. Similarly, $l_{k'} = (x_{k'}, y_{k'}) \in L'$ represents the location of the k -th delivery point.

The parcel weight of the k -th pick-up point is set to be r_k , while the parcel weight of the k -th delivery point is set to be $r'_{k'}$. In particular, the value of $r'_{k'}$ is negative and r'_k is the inverse number of r_k . The reason that it is negative is to indicate that the AAV unloads the parcel with the weight of r_k at the corresponding position of the k -th delivery point. We consider that the AAV has obtained the locations of all the pick-up points and delivery points at the beginning of the mission. Finally, the size of the

area served by the cargo AAV is $U \times U$, which means that the AAV cannot exceed this area when flying.

A. Path Loss Model Between AAV and GBSs

In AAV-enabled communication systems, the path loss model describes the attenuation or loss encountered by the signal during the propagation process. This model is often used to predict the signal strength between the AAV and the GBS under certain conditions. One of the common path loss models is the free space path loss model, also known as the Friis formula. Assuming that there are no interference sources and reflections, such a model is suitable for an open free space environment. However, in the actual environment, there are usually many factors that affect path loss, such as reflection, refraction, and scattering, etc. Therefore, more complex models take these factors into account. In this paper, we use the log-distance model and 3rd Generation Partnership Project (3GPP) specification [32], [33], [34], namely,

$$PL(t) = \begin{cases} 28 + 22 \log_{10}(d_{ub}^{mb}(t)) + 20 \log_{10}(F_c), & \text{LoS,} \\ -17.5 + (46 - 7 \log_{10}(H_f)) \log_{10}(d_{ub}^{mb}(t)) \\ \quad + 20 \log_{10}\left(\frac{40\pi F_c}{3}\right), & \text{NLoS.} \end{cases} \quad (1)$$

where F_c represents the carrier frequency, $d_{ub}^{mb}(t) = \sqrt{\|L_b^{mb} - \mathbf{q}_u(t)\|^2 + (H_f - H_{bs})^2}$ is the euclidean distance between AAV and m_b -th BS at time t , H_{bs} represents the height of the GBS's antenna and all GBSs have the same antenna height, and the position of the m_b -th GBS is represented by $L_b^{mb} = (x_b^{mb}, y_b^{mb})$. From formula (1), we can see that the path loss is related to whether there is LoS channel between AAV and GBS, while the LoS channel depends on the relative position between the AAV, GBS, and ground buildings. This can be determined by checking whether the communication link between AAV and GBS is blocked by the building. In our considered system model, there are the total of M_b GBSs in the area served by the cargo AAV.

B. Communication Model Between AAV and GBSs

According to the standard 3GPP specification [34], [35], [36], every GBS is expected to produce three signal cells. Therefore, there are a total of $G = 3 \times M_b$ cellular cells in the AAV service area. We stipulate that the AAV must establish a stable communication link with only one cell, and the scheme for AAV communicating with cellular cells is the time division multiple access (TDMA) technology. Let $g_c(t)$ represent the cell establishing a communication connection with the cargo AAV at time t , and the signal power $P_g(t)$ received by the cargo AAV from g -th cell at time t is expressed as

$$P_g(t) = P_b^g |h_g(t)|^2, g = \{1, 2, \dots, G\} \quad (2)$$

where P_b^g is the transmit power of cell g and is assumed to be a constant [25], $h_g(t)$ represents the baseband equivalent channel from cell m to the AAV at time t and the value of $h_g(t)$ depends on the antenna gain $A_g(t)$, large-scale fading $\beta_g(t)$ including the path loss $PL(t)$ given in formula (1), and small-scale fading

$\tilde{h}_g(t)$. Moreover, $A_g(t)$ is dependent on antenna tilt angle, carrier frequency, and wavelength and other parameters.

The AAV communicate with no more than one cell, so the signals received by the cargo AAV from other GBSs are viewed as interference. Generally, one can use signal-to-interference ratio (SIR) to evaluate the communication quality of the cargo AAV. Then, the SIR $\psi_g(t)$ received by the cargo AAV from cell g at time t is given as

$$\psi_g(t) = \frac{P_g(t)}{\sum_{m=1, m \neq g}^G P_m(t)}. \quad (3)$$

In our considered system, the AAV needs to establish a good communication connection with GBSs as much as possible, so the best strategy is to select the GBS with the largest SIR. Therefore, the scheduling strategy becomes

$$\psi_{g_c(t)}(t) \geq \psi_g(t), \forall g \in \{1, 2, \dots, G\}, g \neq g_c(t). \quad (4)$$

$$g_c(t) \in \{1, 2, \dots, G\}. \quad (5)$$

In addition, a threshold ϕ_{th} is defined to evaluate the strength of the signal received by the AAV. If $\psi_{g_c(t)}(t) \geq \phi_{th}$, it means that the AAV is in communication with the GBSs at time t , otherwise the system outage P_{out} happens and its corresponding mathematical expression is

$$P_{out}(\mathbf{q}_u(t)) = \Pr\{\psi_{g_c(t)}(t) < \phi_{th}\}. \quad (6)$$

where $\Pr\{\cdot\}$ denotes the probability operation. If the total time required for the AAV to deliver all the parcels is T_d , the total outage time T_{out} of the AAV during the entire mission can be calculated as follows:

$$T_{out}(\mathbf{q}_u(t)) = \int_0^{T_d} P_{out}(\mathbf{q}_u(t)) dt. \quad (7)$$

C. P&D Model of AAV

The P&D order of the cargo AAV is defined as $D = \{d_0, d_1, d_2, \dots, d_{2K}\}$. For any element $d_m \in D, m \neq 0$, the possible values of d_m are within the set $\{1, 2, \dots, K\}$ and the set $\{1', 2', \dots, K'\}$. If $d_m = i$ and $i = 1, \dots, K$, it means that the destination of the AAV's m -th pick-up delivery service is the location of the i -th pick-up point, while $d_m = j'$ and $j = 1, \dots, K$ means that the destination of the AAV's m -th pick-up delivery service is the location of the j -th delivery point. The hangar label is 0 and $d_0 = 0$ indicates that the takeoff point of the AAV should be a hangar. Generally, the parcels must be picked up before being sent to their corresponding users. Therefore, the P&D order D of the cargo AAV meets the following constraints

$$d_i \neq d'_j, \quad \forall d_i, d_j \in D, i < j. \quad (8)$$

In addition, the overall user satisfaction is also an important performance metric, so we use S_{us} to represent overall user satisfaction defined as the weighted sum of the time that it takes for AAV to deliver the corresponding parcels to each user, and S_{us} can be calculated as follows:

$$S_{us} = \sum_{k=1}^K s_k t_k. \quad (9)$$

where t_k represents the time taken by AAV to deliver the parcel to the k -th delivery point, and s_k represents the satisfaction factor of user k . To avoid excessive complexity of the system model, we assume that the satisfaction factors of each user are the same, i.e., $s_i = s_j = s, \forall i, j \in \{1, 2, \dots, K\}$.

D. Energy Consumption Model of AAV

In the AAV pick-up and delivery system studied in this paper, the battery energy of the AAV is mainly used for propulsion energy consumption generated by maintaining flight and communication energy consumption generated by exchanging information with GBSs. According to literature [32], the communication energy consumption of AAV is far less than the propulsion energy consumption. Therefore, in this paper, we ignore the communication energy consumption and only consider the AAV propulsion energy consumption.

The propulsion energy consumption of AAVs is mainly composed of the energy consumption during climbing and descending stages and the energy consumption during horizontal flight. According to literature [37], the propulsion power model of the rotary-wing AAV when flying horizontally is given

$$P_h(V_{uav}) = \underbrace{P_0 \left(1 + \frac{3V_{uav}^2}{U_{tip}^2}\right)}_{\text{blade profile}} + \underbrace{P_i \left(\sqrt{1 + \frac{V_{uav}^4}{4v_0^4}} - \frac{V_{uav}^2}{2v_0^2}\right)^{1/2}}_{\text{induced}} + \underbrace{\frac{1}{2}d_0\rho sAV_{uav}^3}_{\text{parasite}}. \quad (10)$$

Note that the relevant parameters in (10) can be referred to Table I given in [37]. One can clearly see that the flight energy consumption of the AAV is mainly composed of three parts including blade profile, induced, and parasite power. However, in [37], the weight of the AAV is unchanged, while the AAV is constantly picking up or delivering parcels, resulting in changes in the weight of the AAV in our considered scenario. According to [37], parameters P_i and v_0 in (10) are related to the weight $W_u(t)$ of the AAV, and they are obtained as follows:

$$P_i = (1 + k) \frac{W_u(t)^{3/2}}{\sqrt{2\rho A}}. \quad (11)$$

$$v_0 = \sqrt{\frac{W_u(t)}{2\rho A}}. \quad (12)$$

In addition, the propulsion power model of the AAV in the climbing and descending stages is [38]

$$P_v(H_f(t)) = W(t)\dot{H}_f(t), \forall \dot{H}_u(t) > 0. \quad (13)$$

It can be seen from (10) and (13) that the AAV does not consume energy in the descending stage, and the energy consumption in the climbing stage is only related to the weight and climbing height of the AAV. However, the energy consumption in horizontal flight is not only related to the weight of the AAV, but also related to the flight speed of the AAV. Therefore, to reduce the energy consumption of AAV, it is necessary to optimize the P&D order and flight speed of AAV.

Finally, the energy consumption E_f of the AAV can be written as

$$E_f = \underbrace{\int_0^{T_d} P_h(V_{uav}(t), W(t))dt}_{E_h(T_d)} + \underbrace{\int_0^{T_d} P_v(H_f(t))dt}_{E_v(T_d)}. \quad (14)$$

where $E_h(T_d)$ indicates the energy consumption of horizontal flight, and $E_v(T_d)$ indicates the energy consumption of vertical flight.

III. PROBLEM FORMULATION

In this paper, the motivation behind optimizing the P&D sequence of the cargo AAV and adjusting its flight strategy is: 1) minimizing the total energy consumption of the cargo AAV; 2) minimizing the weighted total time required for the cargo AAV to deliver corresponding packages to each user; 3) minimizing the total communication outage of the cargo AAV. Note that there is a tradeoff between these three performance indicators. Thus, we set three different weight factors μ_1 , μ_2 , and μ_3 , and these three weights can be adjusted based on the practical application or the user requirements. Therefore, our proposed optimization problem can be expressed as

$$(P0) : \min_{\mathbf{q}_u(t), D, g_c(t), V_{uav}} \mu_1 E_f + \mu_2 S_{us} + \mu_3 \kappa T_{out} \quad (15)$$

$$\text{s.t. } 0 \leq \|\dot{\mathbf{q}}_u(t)\| \leq V_m, \quad (15a)$$

$$\mathbf{q}_u(0) = l_n, \quad (15b)$$

$$\mathbf{q}_u(T_d) = l_{k'}, \exists k, 1 \leq k \leq K, \quad (15c)$$

$$\mathbf{q}_u(t) = l_k, \exists t \in [0, T_d], \forall l_k \in L, \quad (15d)$$

$$\mathbf{q}_u(t) = l_{k'}, \exists t \in [0, T_d], \forall l_{k'} \in L', \quad (15e)$$

$$0 \leq x_u(t) \leq U, \forall t \in [0, T_d], \quad (15f)$$

$$0 \leq y_u(t) \leq U, \forall t \in [0, T_d], \quad (15g)$$

$$0 \leq H_f(t) \leq H_c, \forall t \in [0, T_d], \quad (15h)$$

$$(5), (8).$$

In problem (P0), we formulate the objective function (15) as a weighted sum of the energy consumption, the communication outage time, and the overall user satisfaction. In (15), μ_1 , μ_2 , and μ_3 represent the non-negative weight factors. In addition, we introduce the balance factor κ because T_{out} is very small compared to E_f and S_{us} [39]. (15a) is the constraint of the cargo AAV and (15b)–(15h) are the flight position constraints of the AAV, where (15b) indicates that the takeoff point of the AAV should be located at the hangar, (15c) means that the cargo AAV should be located at a delivery point when the P&D task is completed, (15d) and (15f) means that regardless of which pick-up point or delivery point, the AAV always arrives at the position of the point at a certain moment, and (15f) and (15g) show that the AAV cannot fly outside the target area. (15h) represents the AAV flight height constraint.

One can clearly see that the above optimization problem is non-convex because it has nonlinear constraints, and E_f and

T_{out} in (15) are quite complex with regard to the AAV's trajectory $\mathbf{q}_u(t)$, the AAV's P&D sequence D , and communication scheduling $g_c(t)$, so it is very difficult to derive E_f and T_{out} mathematically. Therefore, it is difficult to solve the problem through traditional convex optimization methods, and due to the multiple pick-up and delivery points, it is also difficult to directly solve the problem through DRL.

IV. THE PDU STRATEGY FRAMEWORK

To solve problem (P0). We propose a PDU algorithm framework in this section, which divides problem (P0) into two sub-problems to solve, namely: 1) optimization the P&D strategy for the cargo AAV; 2) the trajectory optimization of the cargo AAV. Our idea is to first solve the P&D order and speed of the cargo AAV. Then, the cargo AAV fly in this pick-up and delivery strategy, and optimizes its trajectory to achieve a better communication quality.

A. Optimization of P&D Strategy with SA

The P&D problem of the AAV is similar to the vehicle routing problem (VRP), but it is different from the VRP in that the AAV need strictly comply with the rules of first pick-up and then delivery, i.e., there are some constraints on the flight order of the AAV. Then, the optimization problem can be formulated as

$$(P1) : \min_{D, V_{\text{uav}}} \mu_1 E_f + \mu_2 S_{\text{us}}. \quad (16)$$

The AAV is assumed to fly in a straight line when solving (P1). As we can see from the previous discussion, both vertical and horizontal flight of the cargo AAV consume energy and time. Therefore, to solve the problem (P1), it is necessary to optimize the flight strategy of the vertical flight stage and the horizontal flight stage. Since the vertical flight trajectory and horizontal flight trajectory of the AAV do not affect each other, the flight strategies during vertical flight and horizontal flight can be optimized respectively. In this paper, we use the SA algorithm to solve (P1), and the specific methods are shown as follows

- *Vertical flight strategy optimization:* It can be seen from the vertical propulsion energy consumption model of AAV in Section II that the vertical flight energy consumption of AAV is only related to the height difference caused by vertical flight. In this paper, we do not optimize the altitude, i.e., the altitude difference of the AAV in vertical flight is constant. As can be seen from (9), the overall user satisfaction is related to the delivery time of the parcels, so to minimize the objective function (16), the AAV must fly at the maximum flight speed V_m during the vertical flight stage. Then, the energy consumption $E_v(d_i, d_j)$ and the overall user satisfaction $S_v(d_i, d_j)$ from point d_i to point d_j during vertical flight can be expressed as

$$E_v(d_i, d_j) = \int_0^{\frac{Hc}{V_m}} P_v(H_f(t)) dt = W(t) \times H_c. \quad (17)$$

$$S_v(d_i, d_j) = 2 \times \frac{Hc}{V_m} \times \sum_{m=1}^K c_m. \quad (18)$$

where c_m is an indicator variable and its value is $c_m = 0$ when the cargo AAV delivers the parcels to the m -th delivery point, otherwise $c_m = s_m$.

- *Horizontal flight strategy optimization:* Similarly, the AAV is assumed to fly in a straight line when solving (P1), so the energy consumption $E_h(d_i, d_j)$ and the overall user satisfaction $S_h(d_i, d_j)$ from point d_i to point d_j during horizontal flight can be expressed as

$$E_h(d_i, d_j) = \frac{\|l_{d_i} - l_{d_j}\|}{V_{\text{uav}}(t)} \times P_h(V_{\text{uav}}(t), W(t)). \quad (19)$$

$$S_h(d_i, d_j) = \frac{\|l_{d_i} - l_{d_j}\|}{V_{\text{uav}}(t)} \times \sum_{m=1}^K c_m. \quad (20)$$

where $\|l_{d_i} - l_{d_j}\|$ represents the euclidean distance between point d_i and point d_j .

Therefore, the total energy consumption $E_f(d_i, d_j)$ of the AAV and the overall user satisfaction $S_{\text{us}}(d_i, d_j)$ from point d_i to point d_j are calculated as follows

$$\begin{aligned} E_f(d_i, d_j) &= E_h(d_i, d_j) + E_v(d_i, d_j) \\ &= W(t) \times H_c + \frac{\|l_{d_i} - l_{d_j}\|}{V_{\text{uav}}(t)} \times P_h(V_{\text{uav}}(t), W(t)). \end{aligned} \quad (21)$$

$$\begin{aligned} S_{\text{us}}(d_i, d_j) &= S_h(d_i, d_j) + S_v(d_i, d_j) \\ &= \left(\frac{\|l_{d_i} - l_{d_j}\|}{V_{\text{uav}}(t)} + 2 \times \frac{Hc}{V_m} \right) \times \sum_{m=1}^K c_m. \end{aligned} \quad (22)$$

We define the optimal flight speed of the AAV from point d_i to point d_j as V_{d_i, d_j}^* , and the calculation formula is

$$V_{d_i, d_j}^* = \arg \min_{0 \leq V_{\text{uav}}(t) \leq V_m} \mu_1 E_f(d_i, d_{i+1}) + \mu_2 S_{\text{us}}(d_i, d_{i+1}). \quad (23)$$

After optimizing the AAV's speed, (P1) can be rewritten as

$$(P2) : \min_D \sum_{i=0}^{2K-1} \mu_1 E_f(d_i, d_{i+1}) + \mu_2 S_{\text{us}}(d_i, d_{i+1}). \quad (24)$$

In (P2), the objective function (24) can be minimized by selecting the optimal P&D order and the speed of AAV. It can be found that (P2) is similar to VRP and can also be solved by the commonly used algorithm for solving VRP, including heuristic algorithm, rapidly exploring random tree (RRT) algorithm and dynamic programming (DP) algorithm. SA algorithm is chosen for the following reasons: First, SA algorithm can effectively explore a large solution space and accept suboptimal solutions through a probabilistic acceptance criterion to avoid falling into local optimal solutions. This is different from RRT algorithm, which can quickly explore feasible paths but may encounter difficulties in optimizing the complex objectives and constraints of the VRP problem. Second, compared with DP algorithm that relies on exhaustive search techniques, the scalability and computational efficiency of large-scale VRP instances can be challenging. The SA algorithm can effectively balance exploration and utilization through the iterative adjustment solution of temperature parameters and acceptance probability. Therefore,

Algorithm 1: P&D Strategy Optimization Based on SA Algorithm.

```

1: Initialize: initial temperature  $T_0$ , termination
   temperature  $T_f$ , cooling coefficient  $a_c$ , random initialize
   optimal P&D order  $D^*$ . Set the maximum number of
   iterations to  $K_t$ . Set the objective function (24) to  $f(D)$ .
   Set current task iteration number  $i \leftarrow 0$ .
2: while  $i < K_t$  and  $T > T_f$  do
3:   Randomly select a neighborhood solution  $D_i$  at the
      $i$ -th iteration, calculate the increment of the objective
     function (24) as  $\Delta_f = f(D_i) - f(D^*)$  based on
     (21)–(23);
4:   if  $\Delta_f < 0$  then
5:      $D^* = D_i$ ;
6:   else
7:     Generate random numbers  $\xi = U(0, 1)$ ;
8:     if  $\xi < \exp(-\Delta_f/T)$  then
9:        $D^* = D_i$ ;
10:    end if
11:  end if
12:  Reduce temperature  $T = T \times a_c$ ;
13:   $i = i + 1$ ;
14: end while

```

the SA algorithm has significant advantages over RRT algorithm and DP algorithm. The specific solution method can be found in Algorithm 1.

B. Flight Trajectory Design Through DRL

After solving (P1) to obtain the best pick-up and delivery order and the flight speed of the AAV, we need to further optimize the trajectory of the AAV to realize the performance tradeoff in the AAV pick-up and delivery system. However, solving this problem is very challenging for the following aspects: First, the flight environment of AAV is unknown, i.e., it is difficult to obtain a prior channel environment model. Second, even if there is a channel environment model within the AAV flight area, the model is time-varying, and it is difficult to build an accurate radio map based on the channel environment model. Finally, even without considering the time-variability of the channel model, the optimization problem model is very complex and difficult to be solved by convex optimization methods [25].

Unlike traditional optimization methods, which usually require an accurate model of the system or environment, DRL can learn strategies directly from interactions with the environment, without the need for an accurate system model. DRL can gradually optimize learning strategies through a large number of interactions and experiences, so that it has strong adaptability and generalization ability. It is able to continuously improve strategies in real-time interactions, adapt to changes and unknowns in the environment, and thus better cope with complex optimization problems [40]. Therefore, in this paper, we use twin delayed deep deterministic policy gradient (TD3) algorithm in DRL to solve the tradeoff problem in AAV pick-up and delivery system.

When the P&D strategy of the cargo AAV is determined, E_f and S_{us} in (P0) are the parameters related to the total flight time T_d of the AAV. It is easy to see that E_f and S_{us} increase with the increase of the total flight time T_d of the cargo AAV. To make it easier to use TD3 to solve the optimization problem, we simplify the optimization problem to

$$\begin{aligned}
 \text{(P3)} : \quad & \min_{\mathbf{q}_u(t), g_c(t)} T_d + \mu_3 T_{\text{out}} \\
 \text{s.t.} \quad & (15a) - (15h), \\
 & (5), (8).
 \end{aligned} \tag{25}$$

The first step in using TD3 to solve continuous problems is to transfer continuous problems to discrete problems, i.e., reformulating the problem by a Markov decision process (MDP). In a full MDP, the agent, i.e., cargo AAV observes a state S_n , takes an action A_n , receives an immediate reward R_n , and transitions to the next state S_{n+1} at each discrete time step n . For the AAV trajectory optimization, we need to divide the continuous time range $[0, T_d]$ into N_d time periods with a constant length Δ_t , namely,

$$T_d = N_d \Delta_t. \tag{26}$$

Accordingly, all of the continuous time parameters associated with the duration of the AAV flight become discrete, such as the signal power $P_g(t)$ received by the cargo AAV, the antenna gain $A_g(t)$ of cell g , large-scale fading $\beta_g(t)$, small-scale fading $\tilde{h}_g(t)$, and communication scheduling $g_c(t)$. To ensure accuracy, these parameters should be approximately fixed within a time slot, i.e.,

$$\begin{cases} P_g(t + k\Delta_t) = P_g(t), \forall 0 \leq k \leq 1 \\ A_g(t + k\Delta_t) = A_g(t), \forall 0 \leq k \leq 1 \\ \beta_g(t + k\Delta_t) = \beta_g(t), \forall 0 \leq k \leq 1 \\ \tilde{h}_g(t + k\Delta_t) = \tilde{h}_g(t), \forall 0 \leq k \leq 1 \\ g_c(t + k\Delta_t) = g_c(t), \forall 0 \leq k \leq 1 \end{cases}. \tag{27}$$

To ensure that the conditions in (27) holds, the length of the time slot Δ_t must be small enough so that the position of the AAV in any time slot is approximately unchanged. Then, $A_g(t)$, $\beta_g(t)$, $\tilde{h}_g(t)$, and $g_c(t)$ are discretized to A_g^n , β_g^n , $\tilde{h}_g^n$, and g_c^n , respectively. In addition to the above-mentioned parameters being discretized, the trajectory $\mathbf{q}_u(t)$ of the AAV is also discretized to $\mathbf{q}_n$, namely,

$$\begin{cases} \mathbf{q}_{n+1} = \mathbf{q}_n + \Delta_q \tilde{\mathbf{v}}_n \\ x_{n+1} = x_n + \Delta_q \cos(\varphi_n), \\ y_{n+1} = y_n + \Delta_q \sin(\varphi_n) \end{cases}, \tag{28}$$

where Δ_q represents the displacement length of the AAV in one time step, i.e., $\Delta_q = \Delta_t \times V_{\text{uav}}$. x_n and y_n represent the horizontal and vertical coordinates of the position of the AAV at the n -th time step, respectively. φ_n represent the horizontal direction of the AAV in the n -th time step.

Moreover, the expression in (7) becomes

$$T_{\text{out}}(\mathbf{q}_n) \approx \Delta_t \sum_{n=1}^{N_d} P_{\text{out}}(\mathbf{q}_n). \tag{29}$$

As mentioned in Section III, the communication outage probability P_{out} is dependent on the flight path and communication scheduling of the AAV in a very complex form, so it is challenging to derive its closed-form expression directly. To solve this problem, we define an indicator function O used to indicate whether the AAV maintains communication with GBS at the current location $\mathbf{q}_n$

$$O(\mathbf{q}_n, g_c^n) = \begin{cases} 1, & \psi_{g_c^n}[n] < \phi_{th}, \\ 0, & \psi_{g_c^n}[n] \geq \phi_{th}. \end{cases} \quad (30)$$

Then, we have

$$P_{\text{out}}(\mathbf{q}_n) = \Pr \{ \psi_{g_c^n}[n] < \phi_{th} \} = \mathbb{E} [O(\mathbf{q}_n, g_c^n)]. \quad (31)$$

After discretizing the trajectory of the AAV, we utilize the established soft switching mechanism. These mechanisms require regular reporting of Reference Signal Receiving Power (RSRP) and Reference Signal Receiving Quality (RSRQ) values to calculate the SIR between AAV and GBSs for each discrete time interval. According to [25], as the sample size increases, the empirical outage probability $\hat{P}_{\text{out}}(\mathbf{q}_n)$ employed to approximate $P_{\text{out}}(\mathbf{q}_n)$ can be expressed as

$$\hat{P}_{\text{out}}(\mathbf{q}_n) \triangleq \frac{1}{M} \sum_{m=1}^M O_m(\mathbf{q}_n, g_c^n). \quad (32)$$

where M represents the total number of samples, O_m denotes the result of the m -th SIR Sampling. In theory, the more sampling times, i.e., M is large enough, the closer the value of the empirical outage probability $\hat{P}_{\text{out}}$ is to the true value P_{out} . We create a feedforward fully-connected artificial neural network (ANN) with parameters θ , and train it based on P_{out} to learn a model, i.e., the radio map [41]. Then, problem (P3) can be reformulated as

$$\begin{aligned} \text{(P4)} : \quad & \max_{\mathbf{q}_n, g_c^n} -N_d - \mu_3 \sum_{n=1}^{N_d} \hat{P}_{\text{out}}(\mathbf{q}_n) \\ \text{s.t.} \quad & (15a) - (15h), \\ & (5), (8). \end{aligned} \quad (33)$$

Next, to solve the problem (P4), we reformulate the objective function given in problem (P4) as an MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R} \rangle$

- **State $\mathcal{S}$:** As mentioned in Section II, the AAV cannot fly beyond the boundary of its cargo distribution area. The flight status of the AAV at the n -th time step is represented by $S_n = \mathbf{q}_n = (x_n, y_n)$, where the legality condition stipulates that $0 \leq x_n \leq U$ and $0 \leq y_n \leq U, \forall n$.
- **Action $\mathcal{A}$:** The action of the AAV at the n -th time step is denoted by φ_n , and φ_n represents the horizontal flight direction of the AAV in the n -th time slot. It should be noted that the action space of the AAV at any time is continuous, and the range of action φ_n is: $\varphi_n \in (0, 2\pi], \forall n$.
- **Probability $\mathcal{P}$:** The state transition probability related to the strategy of flight is determinately decided by (28).

- **Reward $\mathcal{R}$:** The reward function r_n for AAV moving at n -th step is set to

$$r_n = \begin{cases} R_{\text{rd}}, & \text{if the AAV reaches its destination;} \\ R_{\text{ob}}, & \text{if the AAV flies out of the designated area;} \\ -1 - \mu_3 \hat{P}_{\text{out}}(\mathbf{q}_n) + \lambda \|d_{\text{pre}} - d_{\text{cur}}\|, & \text{otherwise.} \end{cases} \quad (34)$$

where d_{pre} and d_{cur} represent the previous and current relative distance between the AAV and the destination [42], respectively. Specifically, the cargo AAV receives a penalty value of 1 when it flies from any point to the next point. The purpose of this punishment is to avoid the trajectory of cargo AAVs being too circuitous, resulting in high performance parameters of energy consumption and overall user satisfaction. Second, the AAV also receives the penalty $\hat{P}_{\text{out}}(\mathbf{q}_n)$ and its value is dependent on the communication quality of the AAV. During training, the agent selects the optimal trajectory according to the size of μ_3 to minimize the objective function. Finally, λ represents the inducing factor, which is a constant and acts to facilitate the AAV to reach its destination [42].

After transforming the optimized problem into MDP, we can implement the framework of TD3. TD3 uses two Q-networks to calculate the state-action value function (Q-value) to reduce the over-estimation bias of Q-value, i.e., $Q_1(\cdot | \theta^{Q_1})$ and $Q_2(\cdot | \theta^{Q_2})$. In addition, in order to improve the stability and convergence of learning, TD3 introduces two critic target network the algorithm, which is $Q'_1(\cdot | \theta^{Q'_1})$ and $Q'_2(\cdot | \theta^{Q'_2})$, respectively [43].

In the training phase, the agent samples a batch of data $(s, a, r, s', \text{done})$ from the replay buffer, and using the Target Actor network to calculate the action at the state s' , i.e.,

$$a' = \pi_{\phi'}(s'), \quad (35)$$

where $\pi_{\phi'}(\cdot)$ denotes the Target Actor network, and ϕ' represents the parameters of Target Actor network. Based on the target policy smoothing regularization, noise is added to the target action a' :

$$a' = a' + \epsilon, \quad (36)$$

$$\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c). \quad (37)$$

Then, based on the idea of double network, the target value is calculated:

$$y = r + \gamma \min_{i=1,2} Q'_i(s', a' | \theta^{Q'_i}). \quad (38)$$

Next, update the online critic and actor networks using the policy gradient method, which can be expressed as follows:

$$\theta^{Q'_i} \leftarrow \arg \min \mathcal{O}^{-1} \sum (y - Q_i(s, a))^2, i = 1, 2, \quad (39)$$

$$\nabla_{\phi} J(\phi) = \mathcal{O}^{-1} \sum \nabla_a Q_i(s, a) \Big|_{a=\pi_{\phi}(s)} \nabla_{\phi} \pi_{\phi}(s). \quad (40)$$

In particular, the update frequency of the target network is represented by F , and the TD3 algorithm uses the soft update mode to update the target network. The TD3 algorithm assigns a weighted average of the old target network parameters and the

Algorithm 2: Flight Trajectory Design Based on DRL.

- 1: **Initialize:** Initialize critic networks $Q(s, a; \theta^{Q_i})$ with random parameters θ^{Q_1} and θ^{Q_2} , initialize actor network π_ϕ with random parameters ϕ .
- 2: **Initialize:** Initialize target networks $\theta^{Q'_1} \leftarrow \theta^{Q_1}$, $\theta^{Q'_2} \leftarrow \theta^{Q_2}$, $\phi' \leftarrow \phi$.
- 3: **Initialize:** the maximum number of episodes N_t , maximum number of AAV flight steps per episode N_f , weight factor μ_3 , initial exploration ϵ_0 , decaying rate α , reward for reaching destination R_{rd} , penalty for out of bounds R_{ob} , the experience replay buffer $\mathcal{B}$ with capacity C_b , the ANN database D for learning radio map, cargo AAV P&D range d_{pd} .
- 4: **for** $n = 0$ to N_t **do**
- 5: Randomly generate initial location $\mathbf{q}_0 \in \mathcal{S}$ and set the time step $n \leftarrow 0$;
- 6: **repeat**
- 7: Choose an action $a_n = \pi_{\phi'}(s_n) + \sigma\epsilon$, where ϵ is a Gaussian noise and σ is a decay constant;
- 8: The AAV performs flight maneuvers and update the state s_{n+1} , and obtain reward r_n ;
- 9: Store transition (s_n, a_n, r_n, s_{n+1}) in the replay memory $\mathcal{B}$;
- 10: Add the measured data $(\mathbf{q}_{n+1}, \hat{P}_{out}(\mathbf{q}_{n+1}))$ to the ANN database D ;
- 11: Sample random minibatch from D and update the network parameter θ to learn model of radio map;
- 12: **if** $|\mathcal{B}| > 200$ **then**
- 13: Sample the Mini-batch $\mathcal{O}$ from the experience replay memory $\mathcal{B}$ based on the sample strategy;
- 14: $a' \leftarrow \pi_{\phi'}(s_{n+1}) + \epsilon$, $\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c)$;
- 15: $y = r + \gamma \min_{i=1,2} Q'_i(s', a' | \theta^{Q'_i})$, $i = 1, 2$;
- 16: Update Critics by minimizing the loss:
- 17: $\theta^{Q'_i} \leftarrow \arg \min \mathcal{O}^{-1} \sum (y - Q_i(s, a))^2$, $i = 1, 2$;
- 18: Update the actor policy by the sampled policy gradient
- 19: $\nabla_{\phi} J(\phi) = \mathcal{O}^{-1} \sum \nabla_a Q_i(s, a)|_{a=\pi_{\phi}(s)} \nabla_{\phi} \pi_{\phi}(s)$;
- 20: Update the target Actor and Critic network:
- 21: $\theta^{Q'_i} = \tau \theta^{Q_i} + (1 - \tau) \theta^{Q'_i}$, $i = 1, 2$;
- 22: $\phi' = \tau \phi + (1 - \tau) \phi'$;
- 23: **end if**
- 24: Update $n \leftarrow n + 1$;
- 25: **until** $\|\mathbf{q}_n - l_k\| \leq d_{pd}$ or $\mathbf{q}_n \notin \mathcal{S}$ or $n \geq N_f$;
- 26: **end for**

new corresponding network parameters to the target network by introducing a learning rate τ [44]:

$$\theta^{Q'_i} = \tau \theta^{Q_i} + (1 - \tau) \theta^{Q'_i}, i = 1, 2, \quad (41)$$

$$\phi' = \tau \phi + (1 - \tau) \phi'. \quad (42)$$

V. SIMULATION RESULT

In this section, we present our simulation results to prove that our proposed algorithm is feasible and effective.

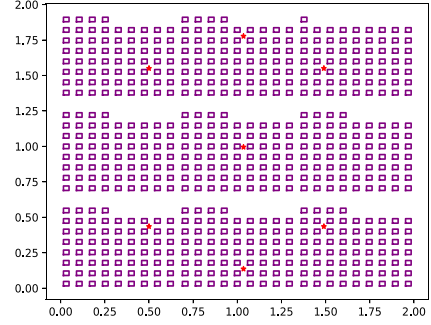


Fig. 2. The urban environment of the AAV flight area.

A. Simulation of Urban Environment

As shown in Fig. 2, the service area of a cargo AAV is defined as a rectangular area with a size of $2 \text{ km} \times 2 \text{ km}$. There are 7 GBSs in the area and the antenna height of the seven GBSs is 25 m, i.e., $M_b = 7$, and $H_{bs} = 25$. These GBSs are represented by a red five-pointed star in Fig. 2, and the antenna tilt angle of all GBSs is 5° . Since there are buildings with different heights in this area, we use a rectangle with a purple border in Fig. 2 to represent the distribution model of the buildings which is created according to the International Telecommunication Union (ITU). Particularly, the environment configuration parameters of buildings can be summarized as

- $\alpha_e = 0.35$: Represents the ratio of the building area within the distribution area compared to the total distribution area, set to 0.35.
- $\beta_e = 320$: Denotes the density of buildings per unit area, specified as 320.
- $\sigma_e = 50$: Indicates that the building heights follow a Rayleigh distribution [25] with a variance of 50. Additionally, the maximum building height is capped at 95 meters.

B. Network Architecture

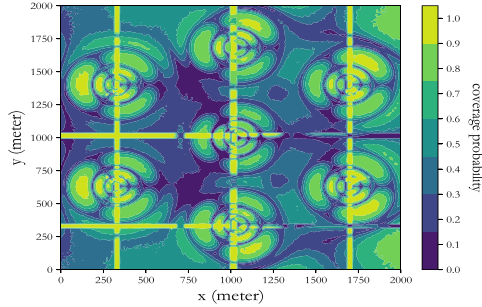
Algorithm 2 applies seven independent ANNs, one of which trains a model based on the data collected by the AAV, i.e., learning a radio map, and the other ANN is used to optimize the flight trajectory of the AAV. Specifically, the number of ANN neurons used to learn the radio map is 512, 256, 128, 64, 32, respectively. The activation for all hidden layers is by Rectified Linear Unit (ReLU), and the ANN is trained with Adam optimizer to minimize the mean square error (MSE) loss, using Python with TensorFlow and Keras [25]. The remaining six ANNs are related to the TD3 algorithm to optimize the flight trajectory of the AAV. The six ANNs are composed of one input layer, three intermediate layers, and one output layer. The number of neurons in the three intermediate layers is 128 [43]. Other parameters used in the simulation are summarized in Table II.

C. Results and Analysis

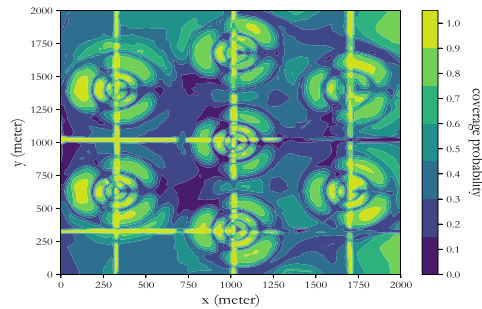
Fig. 3(a) shows the actual radio map of the AAV flight area, in which coverage probability is the complementary probability of the outage probability defined in (6). Note that this radio

TABLE II
PARTIAL SIMULATION PARAMETERS

Simulation parameter	Value
The maximum flight speed of the AAV V_m	50 m/s
The fuselage weight of the AAV W_{aav}	30N
The size of the area U	2
Initial temperature T_0	200
Termination temperature T_f	10
cooling coefficient a_c	0.99
Maximum number of iterations of SA K_t	300
Maximum number of flight steps between any two points N_f	200
Maximum number of training N_t	2000
Time step length Δ_t	0.5 s
cargo AAV P&D range d_{pd}	10
Reward for reaching destination R_{rd}	800
Outbound penalty R_{ob}	-5000
The number of signal measurements after a new state M	1500
Neural coefficient updating frequency of target network F	2
weight factor μ_1	1
weight factor μ_2	1
user satisfaction factor s	300
balance factor κ	100
inducing factor λ	0.5
The learning rate of TD3 algorithm soft update τ	0.005



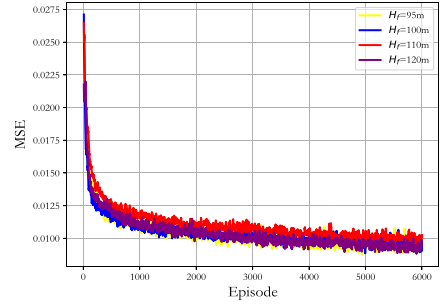
(a) Actual radio map



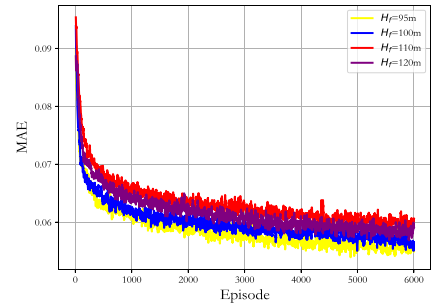
(b) Radio map obtained by learning

Fig. 3. The actual versus learned radio maps.

map is numerically obtained based on the urban environment model and the channel model. However, in the scenario we study, radio map like this one is unknown in advance. From Fig. 3(a), it is evident that the cellular signal coverage pattern exhibits irregularities primarily influenced by the combined impacts of the three-dimensional radiation pattern of GBS's antenna and obstruction from buildings. Notably, within the central region of the area, multiple areas experience weak coverage, where the probability of coverage falls below 20%.



(a) MSE



(b) MAE

Fig. 4. The MSE and MAE versus episode number.

By comparing Figs. 3(a) and (b), it is evident that the radio map obtained by learning is very similar to the actual radio map with only a few differences, which proves that even if there is no channel state information of the AAV's flight area in advance, a radio map with high accuracy can still be learned through the strategy proposed in this paper.

Fig. 4 shows the MSE and mean absolute error (MAE) between the radio map obtained by learning and the actual radio map. It can be seen from Fig. 4 that there is a large error between the learned radio map and the actual radio map when the training episode is small, i.e., the quality of the learned radio map is quite poor at the beginning, but with the increase of the training episode, the error gradually decrease, and the accuracy of the learned radio map is significantly improved. This is because when the training episode is small, the AAV collect less channel environment information, resulting in fewer training data samples. When the training episode is increased, the data collected by the AAV also increase, and the precision of the radio map obtained by training also be improved. In addition, to verify the robustness, we also compare the MSE and MAE between the radio map obtained by learning and the actual radio map at different flight altitudes. Therefore, Figs. 3 and 4 prove that the AAV can still optimize its flight strategy according to the sampled information without obtaining the channel state in advance regardless of the altitude of the AAV flight.

Fig. 5 shows the P&D order when the weight factor $\mu_3 = 0$. In this paper, we use several algorithms for comparison, the specific details of which are as follows:

- User satisfaction-based strategy: The overall user satisfaction is the largest when the AAV flies according to this

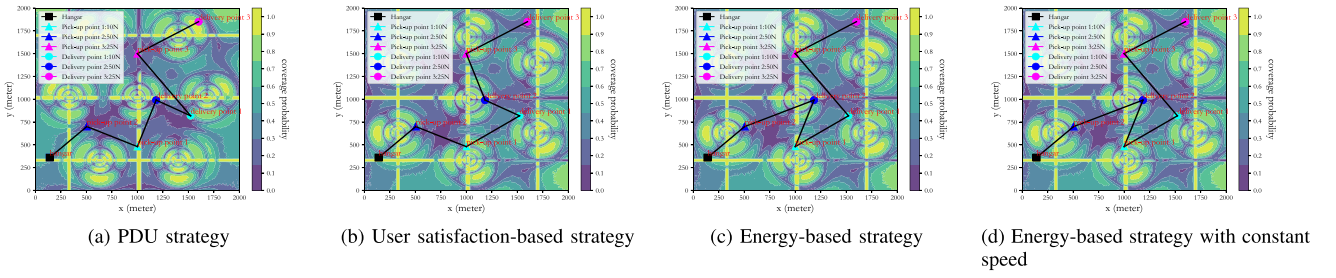


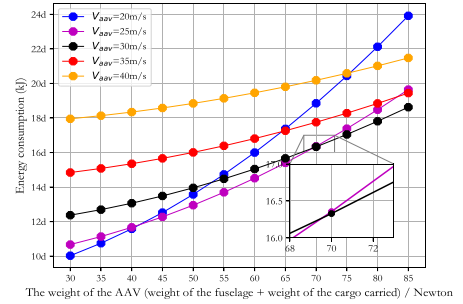
Fig. 5. Trajectory of AAV under different outage thresholds and different schemes.

strategy. Obviously, the AAV should fly at the maximum flight speed V_m , i.e., the AAV speed $V_{uav} = 50$ m/s based on this strategy.

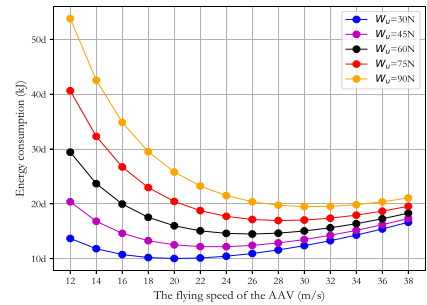
- Energy-based strategy: The energy consumption of AAV is the lowest when it flies according to this strategy, and the flying speed of AAV is variable based on this strategy.
- Energy-based strategy with constant speed: When the speed of the AAV is constant, the energy consumption based on this strategy is the lowest. In this paper, we assume that the AAV's flight speed $V_{uav} = 20$ m/s based on this strategy.

By carefully comparing each subfigure in Fig. 5, it can be found that the length of the AAV flight trajectory obtained by User satisfaction-based strategy is lower than that of the other three strategies. This is because the overall user satisfaction model is related to the flight time, and to improve the overall user satisfaction, it is necessary to reduce the delivery time of each user as much as possible. When based on Energy-based strategy and Energy-based strategy with constant speed, the AAV fly to the corresponding delivery and unload the parcels immediately after obtaining the parcels from the pick-up point, because the larger the payload, the greater the energy consumption of the AAV, and in order to minimize energy consumption, the AAV must carry as little cargo as possible. In particular, the difference between the PDU strategy and User satisfaction-based strategy is that after obtaining the parcels from the pick-up point 1 and pick-up point 2, the PDU strategy first delivers the parcels to the delivery point 2, and then delivers the parcels to the delivery point 1, while User satisfaction-based strategy first delivers the parcels to the delivery point 1, and then delivers the parcels to the delivery point 2. However, by observing the weight of parcels corresponding to delivery point 1 and delivery point 2, it can be found that the weight of parcels corresponding to delivery point 2 is 50N, while the weight of parcels corresponding to delivery point 1 is only 10N. As a result, when User satisfaction-based strategy is adopted, the AAV will carry heavier parcels for a longer flight distance, and the energy consumption of User satisfaction-based strategy is greater than that of the PDU strategy.

To clearly analyze the difference between different strategies theoretically, we first demonstrate the performance improvement brought by optimizing speed., Fig. 6 shows the influence of speed on the performance of the AAV. Fig. 6 shows the relationship between the AAV's power and the payload and the relationship between the AAV's power and the flight speed. As



(a) The relationship between power and payload



(b) The relationship between power and speed

Fig. 6. The flight power of the AAV.

can be seen from Fig. 6(a), when the weight of the AAV is 30N-40N, the AAV consumes the lowest energy when flying at a speed of 20 m/s. When the weight of the AAV is 40N-70N, the AAV consumes the lowest energy when flying at 25 m/s. When the AAV weight is 70N-85N, the AAV consumes the lowest energy when flying at 30 m/s. In particular, when the weight of the AAV is 80N-85N, the AAV consumes the highest energy when flying at a speed of 20 m/s. In addition, it can be seen from Fig. 6(b) that when the weight of the AAV is 30N, the energy consumption of the AAV flying at the speed of 20 m/s is the lowest. When the weight of the AAV is 45N, the energy consumption of the AAV flying at about 22 m/s is the lowest. When the weight of the AAV is 60N, the energy consumption of the AAV flying at about 26 m/s is the lowest. When the weight of the AAV is 75N, the energy consumption of the AAV flying at about 28 m/s is the lowest. When the weight of the AAV is 75N, the energy consumption of the AAV flying at about 28 m/s is the lowest. Therefore, it can be concluded from Fig. 6 that there

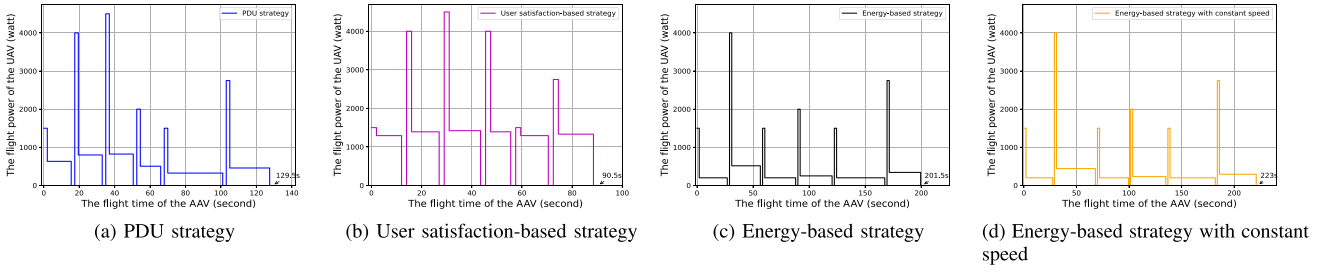
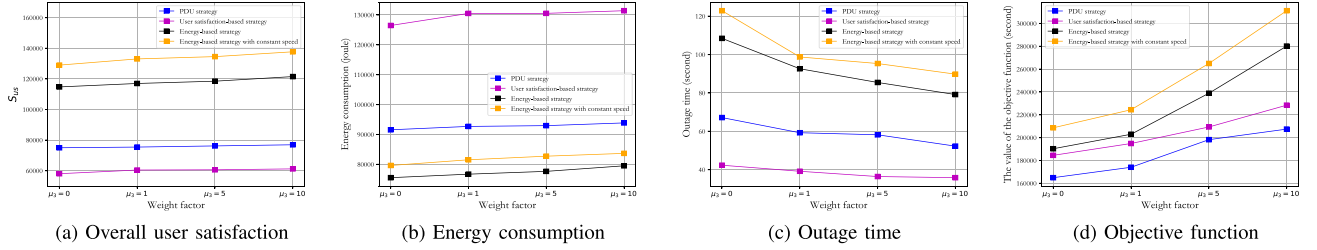
Fig. 7. The flight power versus time when $\mu_3 = 0$.

Fig. 8. Performance parameters.

is a minimum energy consumption speed in the AAV logistics system, and when the AAV flies at this speed, the AAV consumes the lowest energy. Moreover, the larger the weight of the cargo AAV is, the larger the value of the minimum energy consumption speed is, i.e., when the weight of the cargo carried by the AAV is larger, the flight speed of the AAV should be faster.

To more intuitively show the difference caused by different strategies, we show the power versus time and the performance parameters in Figs. 7 and 8. It can be seen from Fig. 7 that the flight time of the AAV based on the User satisfaction-based strategy is the shortest, the PDU strategy is the second, and the flight time of the AAV based on the Energy-based strategy and the Energy-based strategy is the longest. According to (9), parameter S_{us} related to the overall satisfaction of users is related to the delivery time of parcels. Then a longer AAV flight time lead to a decrease in the overall user satisfaction, i.e., the increase of the value of S_{us} . As reflected in Fig. 8(a), the value of S_{us} obtained based on User satisfaction-based strategy is the lowest, followed by PDU strategy and Energy-based strategy, while the value of S_{us} obtained based on Energy-based strategy with constant speed is the highest. In addition, the Y-axis in Fig. 7 represents the power of the AAV, then the area of the closed graph enclosed by the curve in Fig. 7 and the X-axis represents the energy consumption of the AAV. After comparison, it can be found that the area of the closed figure in each subfigure in Fig. 7 is sorted from small to large as follows: Fig. 7(c) < (d) < (a) < (b). This is consistent with the numerical results in Fig. 8(b).

Next, Fig. 8(c) shows that no matter which strategy is based, as the weight factor μ_3 increases, the outage time of the AAV also decrease. This is consistent with the reward mechanism of DRL developed in Algorithm 2. To more intuitively reflect the impact of the change of weight factor μ_3 on the performance of the system, Fig. 9 shows the AAV flight trajectory obtained based on PDU strategy under different weight factors. By comparing

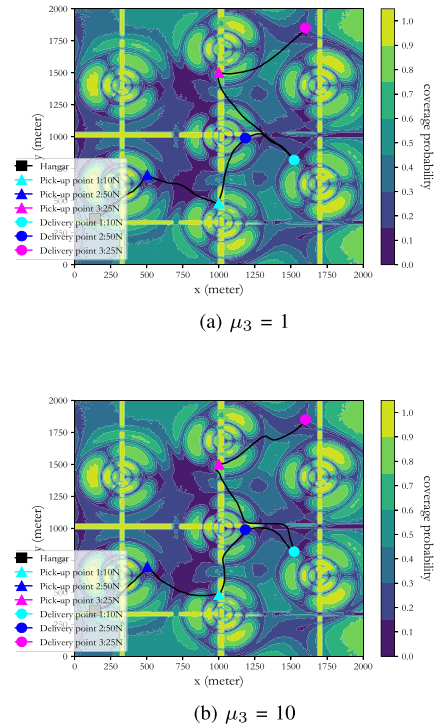
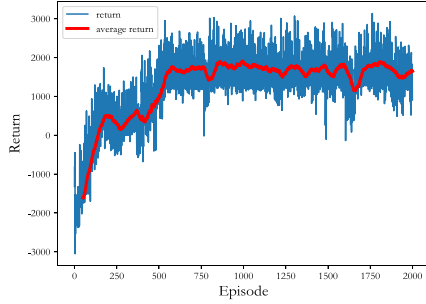
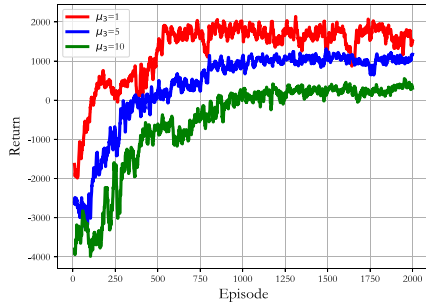


Fig. 9. The flight trajectory of the AAV.

Fig. 9(a) and (b), it can be found that the larger the weight factor μ_3 is, the higher the cellular signal coverage of the area through which the trajectory passes. However, the cost is that the trajectory of the AAV is more circuitous, and the total length of the trajectory increase, which lead to the increase of the energy consumption of the AAV and the decrease of the overall user satisfaction. This is consistent with the trend that the curves of each subgraph in Fig. 8 change with μ_3 .

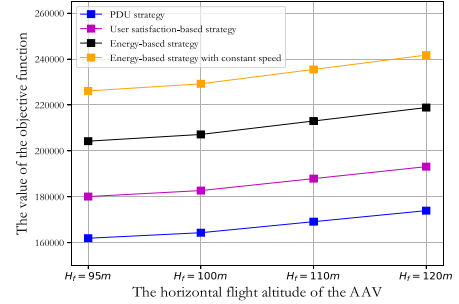
(a) The return per episode when $\mu_3 = 1$ 

(b) The moving average return

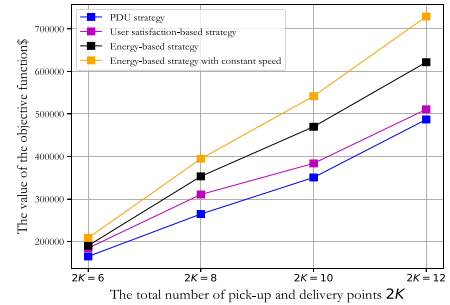
Fig. 10. The return obtained based on the PDU strategy.

Fig. 8(d) shows the value of the objective function obtained by different weights and different strategies. As can be seen from Fig. 8(d), regardless of the weight factor, the PDU strategy proposed in this paper always has the lowest value of the objective function, and its performance is improved by about 5%-30% compared with the comparison strategy. It is proved that PDU strategy is effective in solving the tradeoff problem of energy consumption, overall user satisfaction, and communication quality in the current system model.

To prove the convergence of the PDU strategy, we show the return per episode based on the PDU strategy. The blue and red lines in Fig. 10 represent return per episode when $\mu_3 = 1$ and the moving average return, respectively. As can be seen from Fig. 10(a), in the beginning stage of training, when episode is less than 200, the return based on PDU strategy is very small. However, with the increase of training times, the return shows an increasing trend, although there are some fluctuations. When the episode is greater than 1000, the return hardly changes much. This is because the AAV does not have any prior knowledge about the channel environment. Therefore, in the beginning stage of training, the AAV continue to interact with the environment with a high probability. This exploration stage may cause the agent to experience a lower reward because it has not learned the correct behavior strategy. However, with the increase of training times, the AAV has a better cognition of the environment, so the reward tend to be stable, there are no large fluctuations, and it converge to a stable level. This means that the AAV has learned behavioral strategies to perform well in a given environment. In addition, it can be found from Fig. 10(b) that the larger the value of the weighting factor μ_3 , the smaller the return. This is because



(a) The objective function under different flight altitudes



(b) The objective function under different number of pick-up and delivery points

Fig. 11. The value of the objective function.

according to the reward mechanism of DRL set in Section IV-B, when μ_3 is larger, the possibility of AAV avoiding the weak signal coverage area increase, which lead to the increase of AAV flight trajectory, and ultimately lead to the reduction of return. In short, the PDU strategy is convergent, and the reward mechanism of the DRL is effective.

To prove the robustness of the PDU strategy, we show the value of the objective function under different flight altitudes and different number of pick-up and delivery points in Fig. 11. As can be seen from Fig. 11(a), regardless of the flight height of the AAV, the value of the objective function obtained based on the PDU strategy is always lower than that of the three comparison strategies, and the performance is improved by about 10%-28%. It can also be found that the value of the objective function also increase with the increase of the altitude of the AAV, which is reasonable. As can be seen from (13), the higher the flight altitude of the AAV, the higher the energy consumed in the climbing stage, which also lead to an increase in the time consumed in the climbing and descending stages, thus increasing the value of the objective function. Similarly, Fig. 11(b) shows the performance based on PDU strategy is the best, regardless of the number of pick-up and delivery points. When the number of pick-up and delivery points increases, the AAV needs to carry more parcels and serve more users, which leads to the increase of AAV energy consumption and the decrease of overall user satisfaction, i.e., the reason why the value of objective function increases with the increase of the number of pick-up and delivery points in the system. In conclusion, the PDU strategy proposed in this paper always has the optimal performance in

solving the energy consumption, overall user satisfaction, and communication quality tradeoff problem of the AAV pick-up and delivery system, regardless of the change of AAV flight height or the change of the number of delivery points in the system.

VI. CONCLUSION

In this paper, we studied a AAV pick-up and delivery system. In this system, there is a tradeoff between AAV energy consumption, overall user satisfaction and AAV communication quality. This paper proposed PDU strategy, which is composed of SA algorithm and DRL algorithm. The tradeoff problem is solved through the joint optimization of the P&D order, flight speed, flight trajectory and communication scheduling. Specifically, SA algorithm is used to optimize the P&D order and flight speed of the cargo AAV, while the DRL algorithm is used to optimize the flight trajectory and communication scheduling of the cargo AAV when the channel model is unknown, and learned a local radio map to improve its cognition of the radio environment. Finally, numerical results show that the PDU strategy can learn a radio map with high accuracy from the sampled data of AAV under the premise of unknown channel model, and the performance based on the PDU strategy is improved by about 5%-30% compared with the three comparison strategies, and the PDU strategy is robust and convergent.

Although we study the trade-off between user satisfaction, the energy consumption of the AAV, and the communication quality in the AAV logistics system, the current work does not optimize the flight trajectory of the AAV when flying vertically. At the same time, in order to avoid the collision between the AAV and the building, this paper assumes that the height of the AAV is greater than the height of the building when flying horizontally. This assumption is quite limited, so the 3D trajectory optimization of AAV can be studied in future work while considering the obstacle avoidance problem of AAV. Theoretically, the performance of AAV's three-dimensional trajectory optimization is better than that of two-dimensional trajectory optimization.

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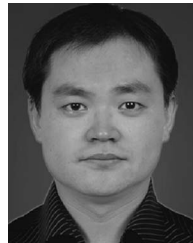
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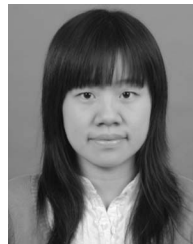
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